manipulation, or bias in how our digital reflections are managed and acted upon.

NOTE

Companies such as Delve.ai (https://delve.ai) market software that offers to build a so-called *digital twin of a customer (DToC)* and a *digital twin of an employee (DToE)*—with the promise of building a digital replica of a person to make predictions about their performance, behavior, and potential.

These predictive digital twins are constantly learning—from your clicks, views, likes, purchases, and even how you move your mouse over the screen. They're not based on models of physics and are more focused on patterns in data, but the principles are the same: observe the past, understand the present, and forecast the future.

1.3.4 Comprehensive digital twin

A *comprehensive digital twin* takes things a step further. It actively simulates different possible futures in addition to monitoring a system and making forecasts. This lets us explore questions like, "What happens if we tweak this parameter?", "What if we push the system beyond its normal limits?" The goal is to understand how a physical system might behave under various conditions, even if we never touch the real thing.

MODELING MEETS REALITY

Simulation and modeling of complex systems isn't new. Engineers and scientists have been doing this for decades, using mathematical models to represent everything from electrical grids to climate systems. Comprehensive digital twins extend traditional modeling by dynamically integrating real-world data, often through techniques such as state estimation or *data assimilation*. Data assimilation is the process of combining real-world observations with mathematical models to create the most accurate possible representation of a system's current state.